POSTER

# Using The HeartRateLab.jl for Normative Evaluation of Heart Rate Variability from Open Datasets

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github.com/abcsds/HeartRateLab.jl


## Abstract

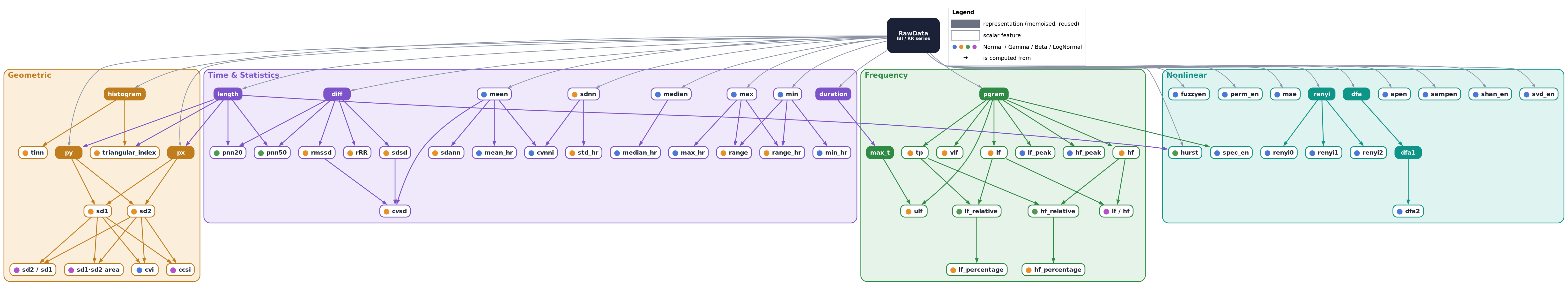
A common method in the evaluation of human biosignals is contrasting a recording against reference datasets of similar experiments. Thanks to the open scientific resources for heart rate analysis, and the resilient network of scientific programming functionality that the Julia community enables, this **normative evaluation** of Heart Rate Variability (HRV) can be done **completely in Julia**. We score each **360-beat** window of a longitudinal self-tracked participant (**P1**) who practises **resonant breathing** against per-feature priors fitted to **61,715 windows from 66 healthy subjects** (PhysioNet *nsrdb+nsr2db*). Against this general population P1's low-frequency band is strongly elevated: **LF/HF** ($z=+2.6$), **LF power** ($z=+2.5$); while short-term vagal tone (**RMSSD** $z=+0.0$, **Mean IBI** $z=+0.1$) stays at the healthy median. Re-scored against *meditators* (*Heart Rate Oscillations during Meditation*, 11 practitioners, themselves only mildly shifted from healthy: pNN50 $z=+0.4$, SDNN $z=+0.8$), P1's elevation roughly halves (**LF/HF** $z=+1.7$): paced resonant breathing places P1 *atypical* for the general population but *meditator-like*. A recording is "normal" only relative to a chosen reference. In this way we demonstrate how correctly implemented open science can answer contextual questions such as: *"Is this new experimental data normal?"*


## Why context matters

We contrast a time series of **Inter-Beat Intervals (IBIs)** against the feature distributions of existing open datasets, emphasising the **contextual nature** of scientific inquiry. This capability is a novel addition implemented specifically for HeartRateLab.jl.

An example contrasting meditation recordings demonstrates current capabilities and illustrates functionality that aids **hypothesis testing**, **feature engineering**, and a deeper understanding of recordings, including use as a **personal daily HRV monitor** that norms your own signals against a healthy population.

## The feature computational graph: 53 features computed from each other

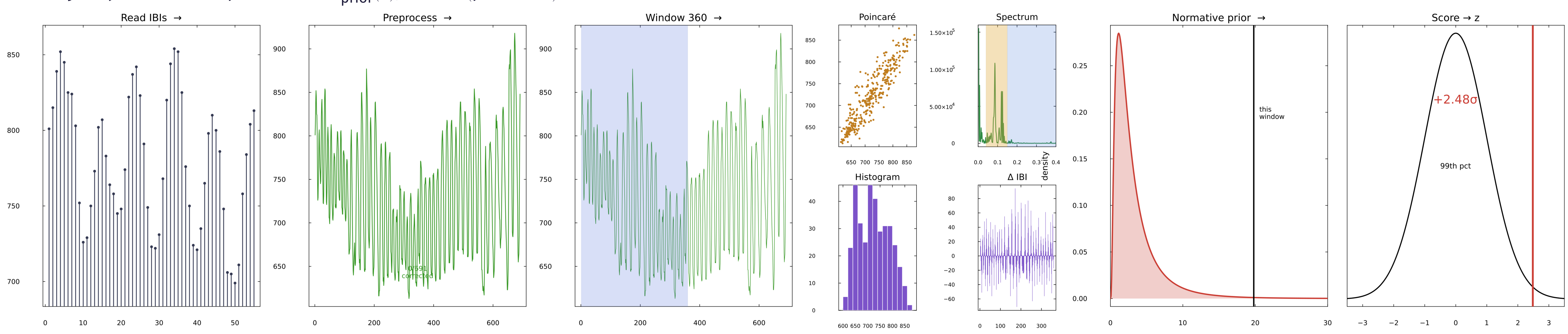



HRV features are **not independent measurements**. One IBI series feeds **11 memoised representations** which feed **53 scalar features**; every arrow is an "is computed from" dependency read from the live `Features.jl` registry. RMSSD, SD1 and SDSD are the *same* quantity up to a constant; LF/HF, LF% and LF-rel are all functions of LF and HF: reporting them as independent evidence multiplies one fact, the **definitional** root of the empirical redundancy the clustermap shows below.

## Method

**Read** IBIs $\to$ **preprocess** (bio-outliers, ectopic & zero beats) $\to$ **53 HRV features** (time / frequency / geometric / nonlinear) $\to$ **window** (360 beats / 120 stride) $\to$ fit a **normative prior** per feature (Normal / Gamma / Beta / LogNormal) by MLE over **61,715 windows from 66 healthy subjects** $\to$ **score**: percentile $= F_{\text{prior}}(x)$, $z=\Phi^{-1}$(percentile).

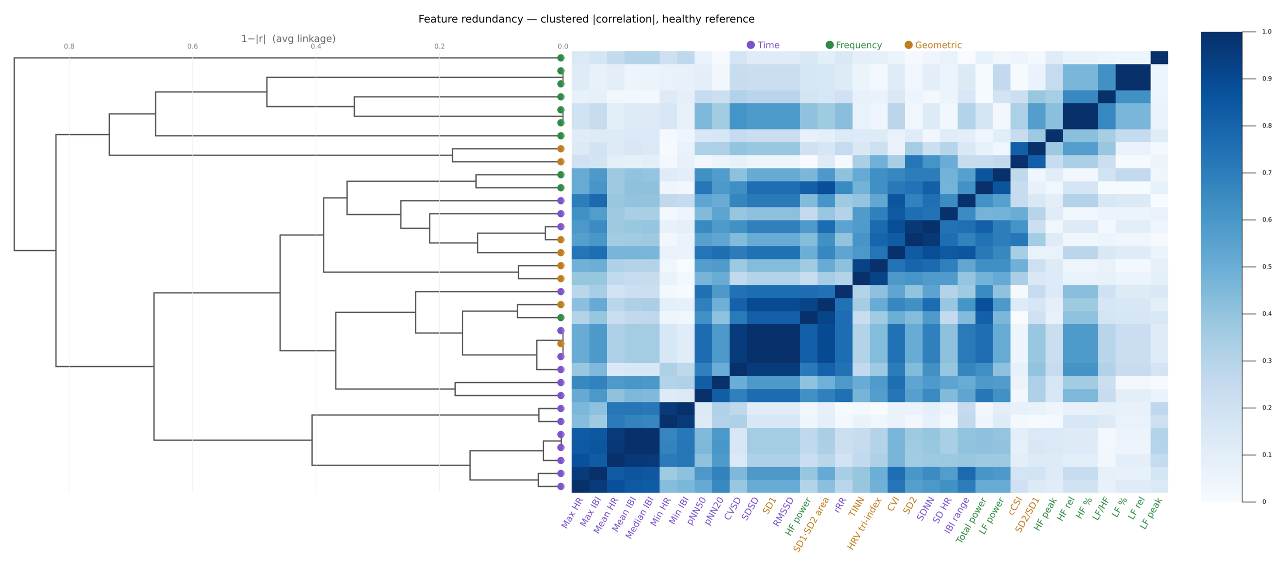
**Scope.** Priors are MLE approximations (KS-rejected at n = 56k); a $z$ is a percentile re-expressed on a standard normal, window-level and descriptive.




One real recording (Participant P1) through every step. **Features** spans all domains (Poincaré, spectrum, histogram, successive differences $\to$ 53 numbers); here the window's LF/HF lands at the 99th percentile of the healthy prior ($z=+2.5$).

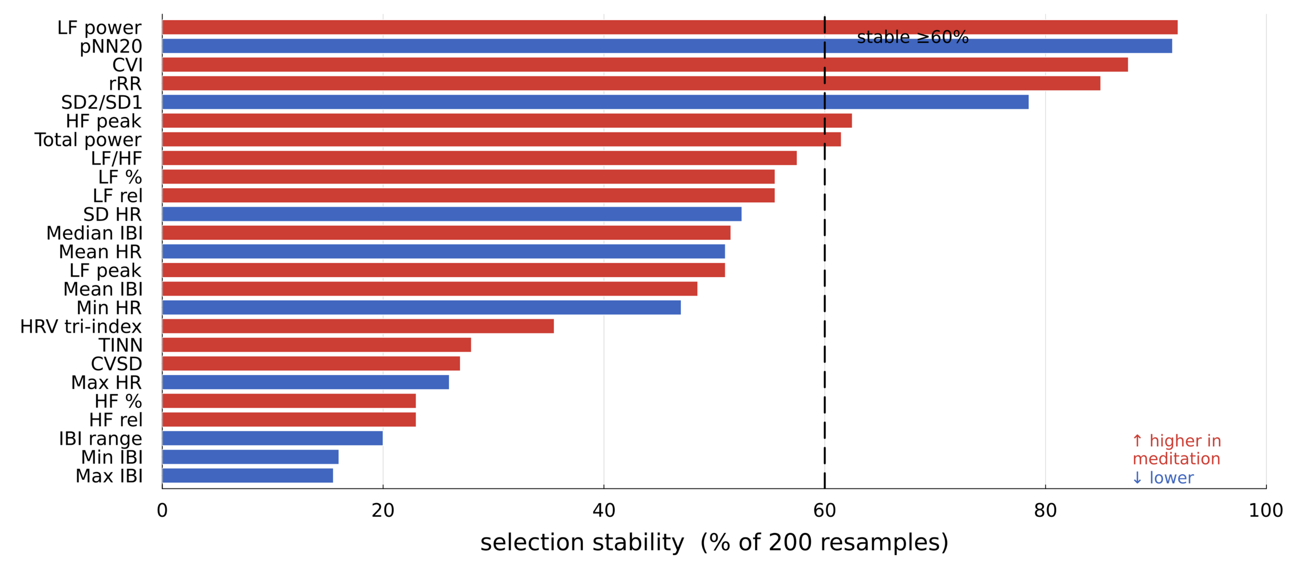
## Results

### HRV features are largely redundant



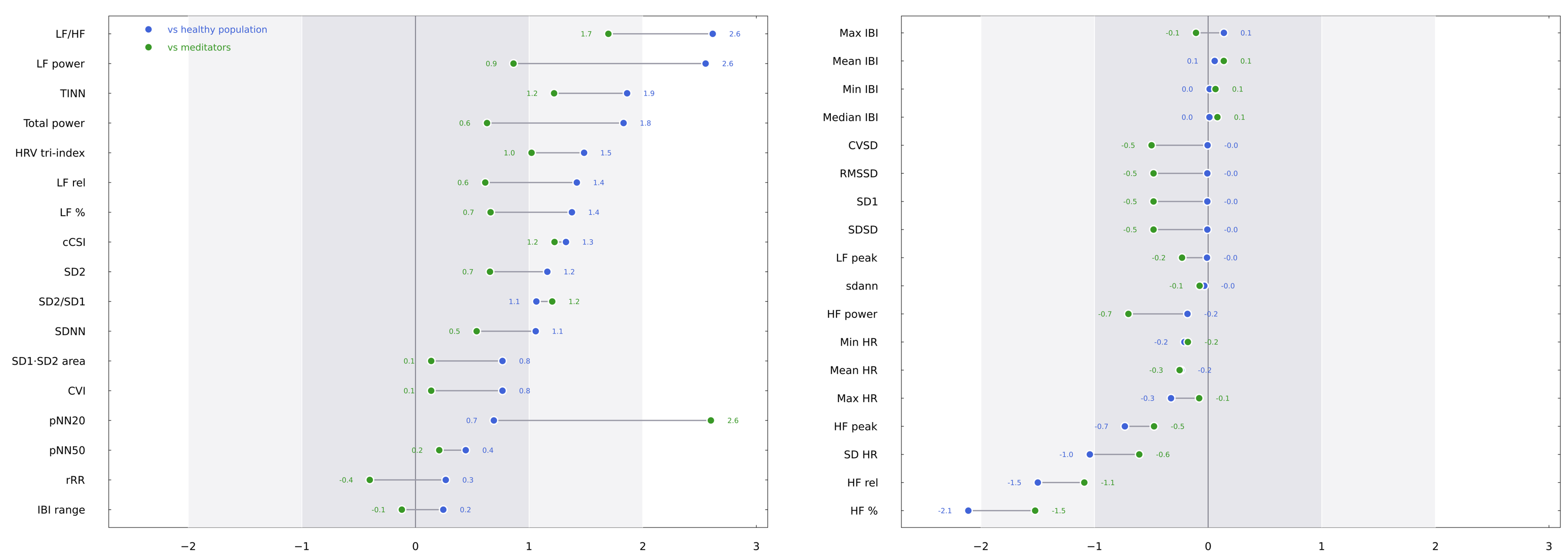
Clustered |correlation| on the healthy reference; the dendrogram groups near-duplicates. Dark blocks: SD1≡RMSSD≡SDSD ($r=1.00$), LF%≡LF-rel, SDNN≈SD2

### Which features distinguish meditation



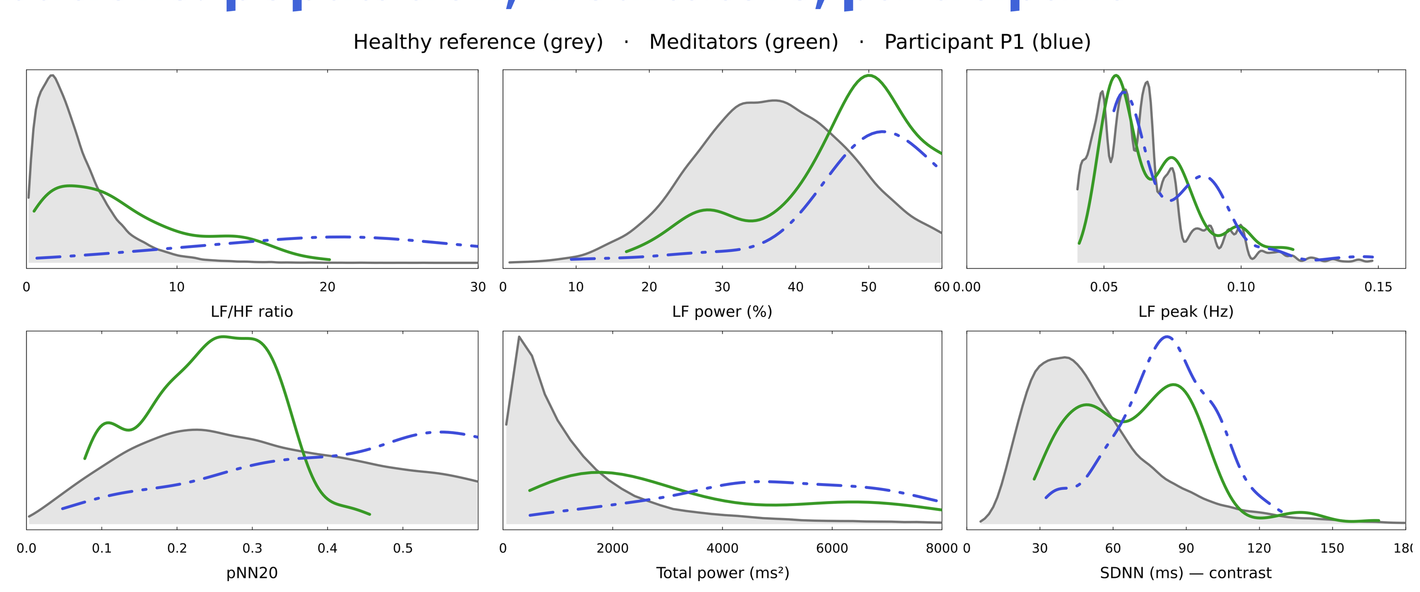

L1-logistic **stability selection** (meditation vs healthy, 200 subject-grouped resamples): the ~0.1 Hz LF-band features are selected most often. Grouped 5-fold CV **AUC** $=0.73\pm0.27$ (only 11 meditators). Features are chosen by the model.

### Participant P1 scored against two references



P1's deviation ($\sigma$-equivalent) from the **healthy population** and from the **meditators**. On the LF-band features P1 is extreme vs the general population (LF/HF $+2.6\sigma$) but close to a typical meditator ($+1.7\sigma$)

### Distributions: population, meditators, participant




Grey = healthy reference (66 subjects); **green** = meditators (11 practitioners); **blue** = P1. Both meditators and P1 shift the LF-band features right of healthy; short-term time-domain spread (SDNN) separates them much less.

## "Is this new experimental data normal?"

HeartRateLab.jl answers it **only in context**. Participant P1 (paced **resonant breathing**) reads as **atypical** against the general population (LF/HF $+2.6\sigma$) yet **meditator-like** against the meditation cohort (LF/HF $+1.7\sigma$). The *same* recording has two verdicts, while the meditators themselves sit $<1\sigma$ from healthy. The contribution is the **framing**: placing one recording in a normative context, not statistical inference on P1.

### Limitations

The "normative" references are convenience datasets (nsrdb/nsr2db), not sampled normal populations; per-feature priors are KS-rejected approximations; windows are correlated (descriptive, not inferential); the meditation cohort is n = 11; **P1 is a single self-tracked participant**. Full detail and the feature table are in the notes handout.

HeartRateLab

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